

Project proposal

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Background

Experimental design plays a fundamental role in agricultural research because the choice of design directly affects the precision of treatment comparisons, statistical power, and the reproducibility of results. Among the designs commonly used in field experiments, split-plot designs are useful when the levels of some treatment factors are more difficult to change during the experiment than those of others. In practice, these experiments are often implemented under a Randomized Complete Block Design (RCBD), although Incomplete Block Designs (IBD) may provide superior local control when environmental variability is substantial. According to Bailey (2008), appropriate blocking can substantially reduce experimental error and improve the efficiency of treatment effect estimation, highlighting the importance of selecting an adequate design structure.

Recent developments in Bayesian statistics offer a flexible framework for evaluating and optimizing experimental designs by incorporating uncertainty and prior information into the analysis. Bayesian approaches have been successfully applied to agricultural field experiments and complex split-plot structures, providing more comprehensive inference than traditional methods (Besag & Higdon, 1999; Robinson et al., 2009). Therefore, this project aims to investigate and compare the performance of split-plot experiments under RCBD and IBD structures within a Bayesian framework. Through simulation studies, the research will evaluate how different design configurations influence estimation precision, uncertainty quantification, and overall experimental efficiency, ultimately providing practical guidelines for improving the planning and analysis of agricultural experiments.

References

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- Robinson, T. J., Anderson-Cook, C. M., & Hamada, M. (2009). Bayesian analysis of split-plot experiments with nonnormal responses. *Technometrics*, 51(1), 56–65.